Binomial coefficients mod p: Kummer, Lucas and valuations

Kevin Hu
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L5-X6 Level 5 Advanced · Number theory · Beyond the textbook

Binomial coefficients and primes digit by digit, covering Legendre's formula as a digit sum, Kummer's carries, Lucas's theorem, the prime factors of central binomial coefficients, and Wolstenholme's congruences modulo p^2 and p^3.
By the end you can compute the exact power of a prime dividing a binomial coefficient by counting carries, and count coefficients with a given valuation; evaluate binomial coefficients modulo a prime with Lucas's theorem and use the theorem in proofs; use Wolstenholme's theorem and the central binomial facts with correct hypotheses.
Prerequisites: Orders, Fermat and Euler (L4-NT1), Quadratic residues (L5-NT1). **Contests:** BMO Round 2, EGMO and IMO selection, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

How many of the binomial coefficients $\binom{2026}{k}$, $0 \le k \le 2026$, are odd? How many are not divisible by 3?

In binary $2026 = 11111101010_2$, and in base 3 this number is 2210001_3. The answers are $2^8 = 256$ and $3 \cdot 3 \cdot 2 \cdot 2 = 36$, which come from one more than each binary digit, multiplied together, and the same in base 3. This unit explains why. The tools are four classical results about binomial coefficients and primes. They are Legendre's formula (1808) in its digit-sum form, Kummer's theorem (1852) that the power of p dividing $\binom{m+n}{m}$ is the number of carries when m and n are added in base p, Lucas's theorem (1878) which computes $\binom{n}{k}$ modulo p digit by digit, and Wolstenholme's theorem (1862) which goes from modulo p to modulo p^3. The standard approach treats $\binom{n}{k}$ as a quotient of factorials and computes one valuation at a time with Legendre's formula. That works for one coefficient, but the method gives no hold on questions about all k at once, or on residues rather than divisibility. We assume L4-NT1 (Legendre's formula, Fermat) and use nothing from L5-NT1 beyond working with inverses modulo p.

Notation. Here p is a prime, $v_p(n)$ is the exponent of p in n, and $s_p(n)$ is the sum of the base-p digits of n. When $n = \sum n_i p^i$ we call n_i the i-th digit of n.

1.1 Legendre's formula and Kummer's theorem

Theorem 1 (Legendre).
$v_p(n!) = \sum_{i \ge 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - s_p(n)}{p-1}$.

Proof. The first form is L4-NT1. For the second, write $n = \sum_j n_j p^j$. Then $\lfloor n/p^i \rfloor = \sum_{j \ge i} n_j p^{j-i}$, and summing over $i \ge 1$ the digit n_j appears with weight $p^{j-1} + \cdots + p + 1 = \frac{p^j - 1}{p-1}$. So the sum is $\sum_j n_j \frac{p^j - 1}{p-1} = \frac{n - s_p(n)}{p-1}$. □

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Theorem 2 (Kummer).
For non-negative integers m and n, $v_p\binom{m+n}{m}$ is the number of carries when m and n are added in base p.

Proof. Let a_i, b_i, r_i be the digits of $m, n, m+n$, and let $c_i \in \{0,1\}$ be the carry into position i, with $c_0 = 0$. The addition algorithm says $a_i + b_i + c_i = r_i + pc_{i+1}$ for every i. Sum over all positions, taking enough of them that the last carry is 0. Writing $C = \sum_i c_i = \sum_i c_{i+1}$ for the number of carries, we get $s_p(m) + s_p(n) + C = s_p(m+n) + pC$, that is $s_p(m) + s_p(n) - s_p(m+n) = (p-1)C$. By Theorem 1,

$$v_p\binom{m+n}{m} = \frac{(m+n-s_p(m+n)) - (m-s_p(m)) - (n-s_p(n))}{p-1} = \frac{s_p(m)+s_p(n)-s_p(m+n)}{p-1} = C. \qquad \square$$

For $\binom{n}{k}$ the two summands are k and $n-k$. For $p=2$ and $\binom{2n}{n}$ every 1 in n produces a carry, so $v_2\binom{2n}{n} = s_2(n)$. For odd p the incoming carry matters, as with $p=5$ and $n = 13 = 23_5$, the units give $3+3=6$, a carry, and then $2+2+1=5$ carries again, so $v_5\binom{26}{13} = 2$ although the digit 2 is less than $\frac{5}{2}$.

Corollary 3.
If $p^e \mid \binom{n}{k}$ then $p^e \le n$. Consequently, for $2 \le k \le n-2$ the number $\binom{n}{k}$ is never a prime power.

Proof. Let $p^L \le n < p^{L+1}$. A carry out of position L when adding k and $n-k$ would make $n \ge p^{L+1}$, so carries occur only out of positions $0, \dots, L-1$, and $e \le v_p\binom{n}{k} \le L$. For the second part, if $\binom{n}{k} = p^e$ with $2 \le k \le n-2$, then $n \ge 4$ and $p^e = \binom{n}{k} \ge \binom{n}{2} = \frac{n(n-1)}{2} > n$, a contradiction. $\square$

The same bound follows from Legendre's formula, since each term $\lfloor \frac{n}{p^i} \rfloor - \lfloor \frac{k}{p^i} \rfloor - \lfloor \frac{n-k}{p^i} \rfloor$ is 0 or 1 and vanishes once $p^i > n$, and Theorem 5 below uses this form.

Example 1.
For how many k with $0 \le k \le 2026$ is $\binom{2026}{k} \equiv 2 \pmod 4$?

Solution.
Computing v_2 of each coefficient with Legendre's formula is out of the question. By Kummer we need exactly one carry when k and $l = 2026 - k$ are added in binary. Use the carry equation $k_i + l_i + c_i = n_i + 2c_{i+1}$ with $n = 2026$. Suppose the only carry is out of position i. Then $c_i = 0$ and $c_{i+1} = 1$, so $k_i + l_i = n_i + 2$, which forces $n_i = 0$ and $k_i = l_i = 1$. At position $i+1$ there is no carry out, so $k_{i+1} + l_{i+1} + 1 = n_{i+1}$, which forces $n_{i+1} = 1$ and $k_{i+1} = l_{i+1} = 0$. At every other position $k_j + l_j = n_j$, so $k_j = 0$ when $n_j = 0$, and k_j is free when $n_j = 1$. Conversely each such choice of k gives exactly one carry.
So the count is (number of positions i with $n_i = 0$ and $n_{i+1} = 1$) times $2^{s_2(n)-1}$. Here $2026 = 11111101010_2$ has ones at positions $10, 9, 8, 7, 6, 5, 3, 1$ and zeros at $4, 2, 0$, and each zero has a 1 directly above the zero. So the answer is $3 \cdot 2^7 = 384$.
Remark. The same method counts k with $v_2\binom{n}{k} = j$ for any small j. Describe the possible carry patterns and count the free digits.

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1.2 Lucas's theorem

Theorem 4 (Lucas).
If $n = \sum n_i p^i$ and $k = \sum k_i p^i$ with digits $0 \le n_i, k_i \le p-1$, then

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p},$$

where $\binom{a}{b} = 0$ when $b > a$. In particular $p \nmid \binom{n}{k}$ if and only if $k_i \le n_i$ for every i, and the number of such k is $\prod_i (n_i + 1)$.

Proof. Since $p \mid \binom{p}{j}$ for $0 < j < p$, we have $(1+x)^p \equiv 1 + x^p$, and by induction $(1+x)^{p^i} \equiv 1 + x^{p^i}$, as polynomials with coefficients modulo p. Hence

$$(1+x)^n = \prod_i \left((1+x)^{p^i}\right)^{n_i} \equiv \prod_i \left(1+x^{p^i}\right)^{n_i} = \prod_i \sum_{j=0}^{n_i} \binom{n_i}{j} x^{jp^i}.$$

Expanding the product, choosing $j = j_i$ from the i-th factor gives the monomial $x^{\sum j_i p^i}$, and since $0 \le j_i \le n_i \le p-1$, base-$p$ representation is unique, so x^k arises only from $j_i = k_i$ for all i, and not at all if some $k_i > n_i$. Comparing coefficients of x^k gives the theorem. □

Lucas gives the residue, and Kummer gives the exact power. They agree on divisibility, since there are no carries in $k + (n-k)$ exactly when every digit of k is at most the corresponding digit of n. Two consequences to remember are $\binom{pn}{pk} \equiv \binom{n}{k} \pmod{p}$, and $\binom{p-1}{k} \equiv (-1)^k \pmod{p}$, the latter because $\binom{p-1}{k} = \prod_{j=1}^{k} \frac{p-j}{j}$.

Example 2.
Let p be an odd prime. Find all $n \ge 0$ such that $\binom{n}{k} \equiv (-1)^k \pmod{p}$ for every $0 \le k \le n$.

Solution.
The condition says in particular that no $\binom{n}{k}$ is divisible by p, and Lucas turns that into a condition on the digits of n, and then one well-chosen k decides the top digit. Small cases first. Here $n = p-1$ works by the remark above, and $n = p^2 - 1$ works by Lucas with both digits $p-1$. Try $n = p^m - 1$ in general. All its m digits are $p-1$, so $\binom{n}{k} \equiv \prod_i \binom{p-1}{k_i} \equiv (-1)^{s_p(k)}$, and $s_p(k) \equiv k \pmod 2$ because p^i is odd. So $n = p^m - 1$ works for every $m \ge 0$.
Conversely, suppose n works and let n have top digit $n_L \ge 1$ in position L. No $\binom{n}{k}$ is divisible by p. If some digit n_i with $i < L$ were less than $p-1$, then $k = (p-1)p^i$ satisfies $k < p^{i+1} \le n$ and has digit $p-1 > n_i$ in position i, so $p \mid \binom{n}{k}$ by Theorem 4, a contradiction. So $n = n_L p^L + (p^L - 1)$. Now take $k = p^L \le n$. By Lucas $\binom{n}{p^L} \equiv \binom{n_L}{1} = n_L$, while $(-1)^{p^L} = -1$, so $n_L \equiv -1 \pmod{p}$, that is $n_L = p-1$ and $n = p^{L+1} - 1$. Together with $n = 0 = p^0 - 1$, the answer is $n = p^m - 1$, $m \ge 0$.
Remark. The middle step is worth remembering on its own. No $\binom{n}{k}$ is divisible by p if and only if $n = ap^m - 1$ with $1 \le a \le p$.

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1.3 Central binomial coefficients

Theorem 5.
Let $n \geq 1$ and let p be prime. Then $v_p\binom{2n}{n} = \sum_{i\geq 1}\left(\left\lfloor \frac{2n}{p^i}\right\rfloor - 2\left\lfloor \frac{n}{p^i}\right\rfloor\right)$, each term is 0 or 1, and the terms vanish once $p^i > 2n$. Consequently:

1. $p^{v_p\binom{2n}{n}} \leq 2n$, so $\binom{2n}{n}$ divides $\operatorname{lcm}(1,2,\ldots,2n)$, and $v_p\binom{2n}{n} \leq 1$ when $p > \sqrt{2n}$.
2. every prime p with $n < p \leq 2n$ divides $\binom{2n}{n}$ exactly once.
3. if $n \geq 3$, no prime p with $\frac{2n}{3} < p \leq n$ divides $\binom{2n}{n}$.

Proof. The formula is Theorem 1 applied to $(2n)!/(n!)^2$, and $\lfloor 2x\rfloor - 2\lfloor x\rfloor \in \{0,1\}$ for real x. Part 1 follows because only the indices with $p^i \leq 2n$ contribute. For part 2, $p^2 \geq 2p > 2n$, so only $i = 1$ counts, and $\lfloor \frac{2n}{p}\rfloor - 2\lfloor \frac{n}{p}\rfloor = 1 - 0$. For part 3, $2 \leq \frac{2n}{p} < 3$ gives $\lfloor \frac{2n}{p}\rfloor - 2\lfloor \frac{n}{p}\rfloor = 2 - 2 = 0$, and $p^2 > 2n$ (from $p > \frac{2n}{3}$ when $n \geq 5$, and by inspection, $p = 3$, when $n = 3, 4$). □

As an illustration, take $n = 30$. The primes $31, 37, 41, 43, 47, 53, 59$ in $(30, 60]$ divide $\binom{60}{30}$ once, and 23 and 29, in $(20, 30]$, do not divide $\binom{60}{30}$. For the small primes count carries in $30 + 30$. In base 2, $30 = 11110_2$ gives $v_2 = s_2(30) = 4$. In bases 3 and 5, $30 = 1010_3 = 110_5$ gives no carries. In base 7, $30 = 42_7$ gives one carry. In bases 11, 13, 17 and 19, 30 is 28_{11}, 24_{13}, $(1, 13)_{17}$ and $(1, 11)_{19}$, giving one, none, one and one carry. Hence

$$\binom{60}{30} = 2^4 \cdot 7 \cdot 11 \cdot 17 \cdot 19 \cdot 31 \cdot 37 \cdot 41 \cdot 43 \cdot 47 \cdot 53 \cdot 59,$$

found without computing the number itself.

These are the facts behind Erdős's proof of Bertrand's postulate, which combines them with $\binom{2n}{n} \geq \frac{4^n}{2n+1}$. They also limit what can be proved, such as whether there are infinitely many n with $\binom{2n}{n}$ coprime to 105 is an open problem of Erdős and Graham, so a contest problem of that shape must have extra structure.

1.4 Wolstenholme's theorem

For rationals r, s whose denominators are not divisible by p, we write $r \equiv s \pmod{p^e}$ when the numerator of $r - s$ in lowest terms is divisible by p^e. These congruences can be added, multiplied, and divided by integers prime to p. Write $H = \sum_{k=1}^{p-1} \frac{1}{k}$.

Theorem 6 (Wolstenholme).
Let $p \geq 5$ be prime. Then (a) $\sum_{k=1}^{p-1} \frac{1}{k^2} \equiv 0 \pmod{p}$, (b) $H \equiv 0 \pmod{p^2}$, and (c) $\binom{2p-1}{p-1} \equiv 1 \pmod{p^3}$, equivalently $\binom{2p}{p} \equiv 2 \pmod{p^3}$.

Proof. (a) As k runs over $1, \ldots, p-1$, so does the residue of $\frac{1}{k}$ modulo p, so the sum is congruent to $\sum k^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod{p}$, using $p \geq 5$ to divide by 6. (b) Pair k with $p - k$, so that $2H = \sum_k \left(\frac{1}{k} + \frac{1}{p-k}\right) = p\sum_k \frac{1}{k(p-k)}$, and $\frac{1}{k(p-k)} \equiv -\frac{1}{k^2} \pmod{p}$, so by (a) the last sum is $\equiv 0 \pmod{p}$ and $2H \equiv 0 \pmod{p^2}$. (c) We have

$$\binom{2p-1}{p-1} = \prod_{k=1}^{p-1} \frac{p+k}{k} = \prod_{k=1}^{p-1}\left(1 + \frac{p}{k}\right) = 1 + pe_1 + p^2e_2 + p^3(\cdots),$$

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where $e_1 = H$, $e_2 = \sum_{i<j} \frac{1}{ij} = \frac{H^2 - \sum 1/k^2}{2}$, and the omitted part has denominator prime to p. By (b), $pe_1 \equiv 0 \pmod{p^3}$, and by (a) and (b), $e_2 \equiv 0 \pmod{p}$. □

State the theorem with care. The theorem needs $p \geq 5$, since $\binom{5}{2} = 10 \not\equiv 1 \pmod{27}$ and $\binom{3}{1} = 3 \not\equiv 1 \pmod{8}$ (Babbage's 1819 congruence modulo p^2 holds for all $p \geq 3$). Whether a composite n can satisfy $\binom{2n-1}{n-1} \equiv 1 \pmod{n^3}$ is open. The primes with the stronger congruence modulo p^4 are called Wolstenholme primes, and only 16843 and 2124679 are known.

Example 3.
Let $p \geq 5$ be prime and a a positive integer. Prove that $\binom{ap}{p} \equiv a \pmod{p^3}$, and find the remainder when $\binom{1001}{7}$ is divided by 343.

Solution.
Lucas only gives $\binom{ap}{p} \equiv \binom{a}{1} = a \pmod{p}$. To get further, write the coefficient as a product as in Theorem 6:

$$\binom{ap}{p} = \prod_{k=1}^{p} \frac{(a-1)p+k}{k} = a \prod_{k=1}^{p-1} \left(1 + \frac{(a-1)p}{k}\right),$$

where the factor $k = p$ is $\frac{ap}{p} = a$. Expanding, the product is $1 + (a-1)pe_1 + (a-1)^2p^2e_2 + p^3(\cdots)$ with the same e_1, e_2 as before, so the product is $\equiv 1 \pmod{p^3}$ by Theorem 6. Hence $\binom{ap}{p} \equiv a \pmod{p^3}$. Since $1001 = 143 \cdot 7$, the remainder of $\binom{1001}{7}$ modulo $7^3 = 343$ is 143.
Remark. More generally $\binom{ap}{bp} \equiv \binom{a}{b} \pmod{p^3}$ for $p \geq 5$ (Ljunggren, 1952), but the proof needs more bookkeeping, and we do not use the result.

Common traps.

- Kummer counts carries in $k + (n-k)$, not in $k + n$, and the carry into a position counts, as in base 5, $13 + 13$ has two carries.
- Lucas's theorem is for a prime modulus only. Modulo 4 the digit rule fails, since $\binom{4}{2} = 6 \equiv 2$, while the base-2 digit product is 0.
- Wolstenholme needs $p \geq 5$. Say which congruence you mean when fractions appear, and check that every denominator is prime to p.
- Legendre's digit-sum form divides by $p - 1$, and for $p = 2$ the formula reads $v_2(n!) = n - s_2(n)$.
- A marker wants the base-p expansions written out and the carry positions identified, not just a final count.

1.5 How to spot this in a contest

1. A prime power dividing a binomial coefficient, a product of consecutive integers, or a quotient of factorials suggests Kummer or Legendre.
2. A binomial coefficient with huge entries modulo a small prime suggests Lucas.
3. Counting entries of Pascal's triangle with a divisibility property suggests digit counts from Lucas and Kummer.
4. Harmonic sums, $\binom{2p}{p}$ or $\binom{ap}{bp}$ modulo p^2 or p^3 suggest Wolstenholme's pairing $k \leftrightarrow p - k$.

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5. $\binom{2n}{n}$, $\operatorname{lcm}(1,\dots,n)$, or primes in $(n,2n]$ suggest Theorem 5.
6. Showing that a quotient of factorials is an integer suggests comparing valuations prime by prime, which reduces to an inequality between floor functions.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Find the smallest positive integer n such that 2026^2 divides $\binom{2n}{n}$. (Note that 1013 is prime.)

Problem 2 ⋆⋆ Let p be prime and $n \geq 1$. Prove that $v_p\left(\binom{p^n}{k}\right) = n - v_p(k)$ for every $1 \leq k \leq p^n$.

Problem 3 ⋆⋆ Let p be prime and let n have base-p digits $n_0, n_1, \dots, n_L$. Prove that

$$\sum_{k=0}^{n} \binom{n}{k}^{p-1} \equiv \prod_{i=0}^{L} (n_i + 1) \pmod{p}.$$

Deduce that if every base-3 digit of n is 0 or 1, then $\binom{2n}{n} \equiv (-1)^t \pmod 3$, where t is the number of digits equal to 1, and that otherwise $3 \mid \binom{2n}{n}$.

Problem 4 ⋆⋆⋆ For a positive integer m let $L_m = \operatorname{lcm}(1,2,\dots,m)$. Prove that $\frac{L_{2n}}{L_n}$ divides $\binom{2n}{n}$ for every $n \geq 1$.

Problem 5 ⋆⋆⋆ For a prime p, write $\sum_{k=1}^{p-1} \frac{1}{k^3} = \frac{a}{b}$ in lowest terms. Find all primes p for which p^2 divides a.

Problem 6 ⋆⋆⋆⋆ Let p be a prime and n a positive integer, and write $n+1 = cp^m$ with $m \geq 0$ and $p \nmid c$. Prove that none of $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$ is divisible by p^2 if and only if $c < p^2$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME II 2006 Problem 3** ⋆
   Write the product as a quotient of factorials and apply Legendre's formula twice.
2. **AIME 1983 Problem 8** ⋆
   For a candidate prime, compare its exponent in the numerator with twice its exponent in each factorial of the denominator.
3. **HMMT February 2015 Guts Round Problem 27** ⋆⋆
   The range is chosen to be 0 to 3^4 - 1. Write a and b in base 3.
4. **IMO 1972 Problem 3** ⋆⋆
   Compare the exponent of each prime on top and bottom, and this reduces to an inequality between floor functions.

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5. **APMO 2006 Problem 3** ⋆⋆⋆
   Express the count with a single binomial coefficient, then expand the binomial coefficient as a product and use Wolstenholme's theorem.

6. **IMO 2019 Problem 4** ⋆⋆⋆
   Compare the power of 2 on both sides using Legendre's formula, then compare sizes.

7. **IMO 1979 Problem 1** ⋆⋆⋆
   Turn the alternating sum into a sum of reciprocals over a range, then pair terms as in the proof of Wolstenholme's theorem.

8. **HMMT February 2022 Algebra and Number Theory Problem 10** ⋆⋆⋆⋆
   Use the digit-sum form of Legendre's formula to read the parity of the exponent from the base-p digits.

9. **USAMO 2010 Problem 5** ⋆⋆⋆⋆
   Use partial fractions, then pair terms whose denominators add up to a multiple of p.

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4 Hints

1. $2026^2 = 4 \cdot 1013^2$. Count carries in $n + n$ in base 1013, remembering the incoming carry.
2. Use $k\binom{p^n}{k} = p^n\binom{p^n-1}{k-1}$, and look at the digits of $p^n - 1$.
3. Fermat's little theorem turns each term into 0 or 1. For the deduction, $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$.
4. $v_p(L_m)$ is the largest e with $p^e \le m$. When does $v_p(L_{2n})$ exceed $v_p(L_n)$, and which term of Theorem 5's sum is then equal to 1?
5. Pair k with $p - k$ as in the proof of Wolstenholme's theorem, and reduce to a sum of fourth powers modulo p. Small primes need separate treatment.
6. By Kummer you need the largest number of carries in $k + (n - k)$ over all k. Show that no carry can start at a position where n has digit $p - 1$, and that once a carry starts the carry can be kept going up to the top digit.

5 Solutions

Problem 1

Find the smallest positive integer n such that 2026^2 divides $\binom{2n}{n}$. (Note that 1013 is prime.)

We have $2026^2 = 2^2 \cdot 1013^2$. By Kummer, $v_2\binom{2n}{n} = s_2(n)$ and $v_{1013}\binom{2n}{n}$ is the number of carries in $n + n$ in base 1013. Write $n = a_1 \cdot 1013 + a_0$ when $n < 1013^2$. A carry out of position 0 happens if and only if $2a_0 \ge 1013$, that is $a_0 \ge 507$. A carry out of position 1 happens if and only if $2a_1 + c_1 \ge 1013$, where c_1 is the carry from position 0. This needs $a_1 \ge 507$, or $a_1 = 506$ and $c_1 = 1$.

If $n < 1013^2$, two carries need both of these, so $a_0 \ge 507$ and $a_1 \ge 506$, giving $n \ge 506 \cdot 1013 + 507 = 513085$. If $n \ge 1013^2$ then $n > 513085$. For $n = 513085$ there are two carries, so $1013^2 \mid \binom{2n}{n}$, and n is odd and greater than 1, so $s_2(n) \ge 2$ and $4 \mid \binom{2n}{n}$. Hence the answer is 513085.

Answer: 513085

Problem 2

Let p be prime and $n \ge 1$. Prove that $v_p\binom{p^n}{k} = n - v_p(k)$ for every $1 \le k \le p^n$.

From $\binom{N}{k} = \frac{N}{k}\binom{N-1}{k-1}$ with $N = p^n$ we get $k\binom{p^n}{k} = p^n\binom{p^n-1}{k-1}$. In base p, the number $p^n - 1$ has n digits, all equal to $p - 1$, and $0 \le k - 1 < p^n$ has at most n digits. So every digit of $k - 1$ is at most the corresponding digit of $p^n - 1$, and by Lucas's theorem $\binom{p^n-1}{k-1} \equiv \prod_i \binom{p-1}{j_i} \not\equiv 0 \pmod p$, where j_i are the digits of $k - 1$ (each factor is ± 1 modulo p). Taking v_p of both sides, $v_p(k) + v_p\binom{p^n}{k} = n + 0$, which is the claim.

(Kummer gives the same result. If $v_p(k) = a$, adding k and $p^n - k$ in base p produces a carry out of each of the positions $a, a + 1, \ldots, n - 1$ and nowhere else.)

Problem 3

Let p be prime and let n have base-p digits $n_0, n_1, \ldots, n_L$. Prove that

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$$\sum_{k=0}^{n} \binom{n}{k}^{p-1} \equiv \prod_{i=0}^{L} (n_i + 1) \pmod{p}.$$

Deduce that if every base-3 digit of n is 0 or 1, then $\binom{2n}{n} \equiv (-1)^t \pmod 3$, where t is the number of digits equal to 1, and that otherwise $3 \mid \binom{2n}{n}$.

By Fermat's little theorem, $\binom{n}{k}^{p-1} \equiv 1 \pmod p$ if $p \nmid \binom{n}{k}$, and $\equiv 0$ otherwise. So the sum is congruent to the number of k in $[0, n]$ with $p \nmid \binom{n}{k}$, which by Lucas's theorem is $\prod_i (n_i + 1)$.

For $p = 3$, the identity $\sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k} = \binom{2n}{n}$ (choose n objects from two groups of n) gives $\binom{2n}{n} = \sum_k \binom{n}{k}^2 \equiv \prod_i (n_i + 1) \pmod 3$. If some digit n_i equals 2, the product contains the factor 3, so $3 \mid \binom{2n}{n}$. If every digit is 0 or 1, the product is $2^t \equiv (-1)^t \pmod 3$. (The first case agrees with Kummer, since a digit 2 gives $2 + 2 \geq 3$, a carry.)

Problem 4

For a positive integer m let $L_m = \operatorname{lcm}(1, 2, \ldots, m)$. Prove that $\frac{L_{2n}}{L_n}$ divides $\binom{2n}{n}$ for every $n \geq 1$.

For a prime p, $v_p(L_m)$ is the largest e with $p^e \leq m$. Fix p and let e be the largest exponent with $p^e \leq n$, so $n < p^{e+1}$. Then $p^{e+2} \geq 2p^{e+1} > 2n$, so $v_p(L_{2n})$ is e or $e + 1$, and $v_p(L_{2n}/L_n) \in \{0, 1\}$. This valuation is 1 exactly when $n < p^{e+1} \leq 2n$.

In that case consider the term $i = e + 1$ in Theorem 5, namely $\left\lfloor \frac{2n}{p^{e+1}} \right\rfloor - 2\left\lfloor \frac{n}{p^{e+1}} \right\rfloor = 1 - 0 = 1$, because $1 \leq \frac{2n}{p^{e+1}} < 2$ and $\frac{n}{p^{e+1}} < 1$. All the other terms are 0 or 1, so $v_p\binom{2n}{n} \geq 1 = v_p(L_{2n}/L_n)$. When $v_p(L_{2n}/L_n) = 0$ there is nothing to prove. As this holds for every prime p, $\frac{L_{2n}}{L_n}$ divides $\binom{2n}{n}$. Together with Theorem 5 this gives $\frac{L_{2n}}{L_n} \mid \binom{2n}{n} \mid L_{2n}$.

Problem 5

For a prime p, write $\sum_{k=1}^{p-1} \frac{1}{k^3} = \frac{a}{b}$ in lowest terms. Find all primes p for which p^2 divides a.

Let $S = \sum_{k=1}^{p-1} \frac{1}{k^3}$. Its denominator divides a product of numbers less than p, so $p^2 \mid a$ if and only if $S \equiv 0 \pmod{p^2}$ in the sense of the lesson. Small cases give the following. For $p = 2$, $S = 1$. For $p = 3$, $S = 1 + \frac{1}{8} = \frac{9}{8}$, and $9 \mid 9$. For $p = 5$, $S = 1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} = \frac{2035}{1728}$ and $2035 = 5 \cdot 11 \cdot 37$, so $25 \nmid a$.

Now let $p \geq 5$. For $1 \leq k \leq p - 1$,

$$\frac{1}{k^3} + \frac{1}{(p-k)^3} = \frac{(p-k)^3 + k^3}{k^3(p-k)^3} = \frac{p\,(p^2 - 3pk + 3k^2)}{k^3(p-k)^3}.$$

Since the numerator has a factor p, only the rest of the fraction modulo p matters modulo p^2, and modulo p we have $p^2 - 3pk + 3k^2 \equiv 3k^2$ and $(p-k)^3 \equiv -k^3$. So the pair contributes $\equiv -\frac{3p}{k^4} \pmod{p^2}$, and summing over all k (each pair is counted twice),

$$2S \equiv -3p \sum_{k=1}^{p-1} \frac{1}{k^4} \pmod{p^2}.$$

As k runs over $1, \ldots, p-1$ so does the residue of $\frac{1}{k}$, so $\sum \frac{1}{k^4} \equiv \sum k^4 \pmod p$, and

$$\sum_{k=1}^{p-1} k^4 = \frac{(p-1)p(2p-1)(3p^2 - 3p - 1)}{30}.$$

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For $p \geq 7$ the prime p does not divide 30, so this is divisible by p, hence $2S \equiv 0 \pmod{p^2}$ and $S \equiv 0 \pmod{p^2}$. (For $p = 5$ the formula gives $354 \equiv 4 \pmod{5}$, so $2S \equiv -12 \cdot 5 \not\equiv 0 \pmod{25}$, which agrees with the direct computation.) Hence $p^2 \mid a$ exactly for $p = 3$ and all primes $p \geq 7$.

Answer: $p = 3$ and every prime $p \geq 7$

Problem 6

Let p be a prime and n a positive integer, and write $n + 1 = cp^m$ with $m \geq 0$ and $p \nmid c$. Prove that none of $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$ is divisible by p^2 if and only if $c < p^2$.

Let n have base-p digits $n_0, \ldots, n_L$ with $n_L \geq 1$, and let t be the number of digits $p - 1$ at the end of n, so $n_0 = \cdots = n_{t-1} = p - 1$ and either $t = L + 1$ or $n_t \leq p - 2$. Then $n + 1 = p^t\left(\lfloor n/p^t \rfloor + 1\right)$, and when $t \leq L$ the last digit of $\lfloor n/p^t \rfloor + 1$ is $n_t + 1 \in [1, p - 1]$. So $m = t$ and $c = \lfloor n/p^t \rfloor + 1$ (and $c = 1$ when $t = L + 1$). By Kummer, we only need to find the largest number M of carries in $k + l$ over all $k + l = n$ with $k, l \geq 0$, and the claim is that $M \leq 1$ if and only if $c < p^2$.

We use the carry equation $k_i + l_i + c_i = n_i + pc_{i+1}$ with $c_0 = 0$. *Upper bound.* If $i < t$ and $c_i = 0$, a carry out of position i needs $k_i + l_i = n_i + p = 2p - 1$, which is impossible as digits are at most $p - 1$. By induction $c_1 = \cdots = c_t = 0$. There is no carry out of position L, since otherwise $n \geq p^{L+1}$. So carries can only leave the positions $t, t + 1, \ldots, L - 1$, and $M \leq \max(L - t, 0)$.

Construction. Suppose $t \leq L - 1$. Take $k_i = p - 1$, $l_i = 0$ for $i < t$ (no carries), then $k_t = p - 1$, $l_t = n_t + 1 \leq p - 1$, which gives $k_t + l_t = n_t + p$ and a carry, then $k_i = p - 1$, $l_i = n_i$ for $t < i < L$, which with the incoming carry gives $n_i + p$ and another carry, and $k_L = 0$, $l_L = n_L - 1$, which with the incoming carry gives n_L. These digits add up to n with $L - t$ carries. Hence $M = L - t$ when $t \leq L$, and $M = 0$ when $t = L + 1$.

Conclusion. If $t = L + 1$, then $c = 1 < p^2$ and $M = 0$. If $t \leq L$, then $\lfloor n/p^t \rfloor$ has $L - t + 1$ digits. When $L - t \leq 1$ the number is less than p^2, so $c \leq p^2$, and $c \neq p^2$ because $p \nmid c$, so $c < p^2$. When $L - t \geq 2$ the number is at least p^2, so $c > p^2$. Therefore $M \leq 1$ if and only if $c < p^2$, as required.

For example, with $p = 2$ the rows with no multiple of 4 are those with $n + 1 = 2^m$ or $3 \cdot 2^m$, while row 4 fails because $\binom{4}{1} = 4$. This extends the remark after Example 2, which is the same argument with $M = 0$.